

Name:

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Roll No:

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- Write your name cleanly in **CAPITAL** letters and roll number inside the boxes.
- Write answers in the space provided only. Maximum marks: 20 Time: 12:00 pm - 13:15 pm.

1. Let $f : (M, d) \rightarrow (N, \rho)$ be an isomorphic (i.e. $\rho(f(x), f(y)) = d(x, y)$) and onto map. Show that f is a homeomorphism.

Given: $f : (M, d) \rightarrow (N, \rho)$ & f is onto
 $\rho(f(x), f(y)) = d(x, y)$

To show: f is one-one.

f is continuous f^{-1} is continuous.

Let $f(x_1) = f(x_2)$

$$\rho(f(x_1), f(x_2)) = d(x_1, x_2)$$

$\parallel$
 0

$$\Rightarrow d(x_1, x_2) = 0$$

$$\Rightarrow x_1 = x_2$$

$\Rightarrow f$ is ~~not~~ one one

f^{-1} exists this implies

$$\rho(x, y) = d(f^{-1}(x), f^{-1}(y))$$

take $\epsilon > 0$

$$\text{let } \delta = \epsilon$$

whenever $d(y, x) < \delta = \epsilon$

$$\rho(f(y), f(x)) = d(y, x) < \epsilon$$

$\Rightarrow f$ is continuous.

take $\epsilon > 0$

$$\text{let } \delta = \epsilon$$

whenever $d(y, x) < \delta = \epsilon$

$$d(f^{-1}(y), f^{-1}(x)) \leq \rho(y, x) < \epsilon$$

$\Rightarrow f^{-1}$ is continuous

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$f : (M, d) \rightarrow (N, \rho)$

is bijective.

f is continuous and f^{-1} is continuous

$\Rightarrow f$ is a homeomorphism

..... Rough Work

f is one one

if set is totally bounded.

f is cont take ~~sequence~~ $d(x, y) < \epsilon \quad \rho(f(x), f(y)) < \epsilon \quad \exists x_1, x_2, x_3, \dots, x_n$

f^{-1} is cont $\rho(f(x), f(y)) < \epsilon \quad d(f^{-1}(x), f^{-1}(y)) < \epsilon$

M is connected

$\Rightarrow$ no A, B disjoint such that.

$x_1, x_2, x_3, \dots, x_n$

Can finite
 Suppose M is at most countable

$(x_1, x_2, \dots, x_n)$

$(x_1, x_2, \dots, x_n)$

A, B disjoint such that.

$A \cup B =$

metric space is disconnected

$\Rightarrow \exists A, B$ s.t.

$A \cup B \neq M, A$

2. Let $(\ell_2, \|\cdot\|_2)$ be the complete metric space where $\ell_2 = \{(x_1, x_2, \dots) : \sum_{n=1}^{\infty} |x_n|^2 < \infty\}$ and the norm is given by $\|x\|_2 = (\sum_{n=1}^{\infty} |x_n|^2)^{\frac{1}{2}}$ for $x = (x_1, x_2, \dots, x_n, \dots)$.

(a) Show that the set $A = \{x \in \ell_2 : |x_n| \leq 1/n \text{ for } n \in \mathbb{N}\}$ is a closed set in ℓ_2 .

(b) Is $B = \{x \in \ell_2 : |x_n| < 1/n \text{ for } n \in \mathbb{N}\}$ an open set? Justify.

(c) Is A complete where A is defined in part (a)?

$$(a) A = \{x \in \ell_2 : |x_n| \leq \frac{1}{n} \text{ for } n \in \mathbb{N}\}$$

for set A to be closed.

if $\{x^{(k)}\} \subset A$ & $x^{(k)} \rightarrow x$ then

$x \in A$. $\rightarrow \{ \text{sequence of sequences} \}$

take $x^{(k)} \subset A$ and $x^{(k)} \rightarrow x$

$$\Rightarrow d(x^{(k)}, x) \rightarrow 0$$

we know convergence implies coordinate convergence

$$|x_i^{(k)} - x_i| \rightarrow 0 \quad \forall i \in \mathbb{N}$$

By triangle inequality

$$|x_i - x_i^{(k)} + x_i^{(k)}| \leq |x_i - x_i^{(k)}| + |x_i^{(k)}|$$

$$\Rightarrow |x_i| \leq |x_i - x_i^{(k)}| + |x_i^{(k)}| \leq |x_i - x_i^{(k)}| + \frac{1}{k}$$

$$\Rightarrow |x_i| \leq |x_i - x_i^{(k)}| + \frac{1}{k}$$

taking $k \rightarrow \infty$

$$|x_i| \leq \frac{1}{i} \quad \forall i \in \mathbb{N}$$

$$\Rightarrow x = (x_1, x_2, \dots) \in A.$$

hence A is closed as all limit points of A lie in A

[2+2+1]

$$(b) B = \{x \in \ell_2 : |x_n| < \frac{1}{n} \text{ for } n \in \mathbb{N}\}$$

no B is not an open set

take $x = (0, 0, 0, \dots)$, $x \in B$

$\varepsilon > 0$, $x \in B$

$$B_\varepsilon(x) = \{y \in \ell_2 : \|y - x\|_2 < \varepsilon\}$$

take by archimedian property, $\exists n \in \mathbb{N}$ s.t.

$$n > \frac{1}{\varepsilon} \quad n \varepsilon > 1$$

$$\Rightarrow \frac{1}{n} > \frac{1}{2n} \quad \Rightarrow \frac{\varepsilon}{2} > \frac{1}{n}$$

$$\text{take } z = (0, 0, \dots, \frac{\varepsilon}{2}, 0, \dots)$$

where all elements are 0 than the n th element.

$$\|z - x\|_2 = \frac{\varepsilon}{2} < \varepsilon$$

$$\|z - x\|_2 = \|(0, 0, \dots, \frac{\varepsilon}{2}, 0, \dots)\|_2 = \left[\left(\frac{\varepsilon}{2}\right)^2\right]^{\frac{1}{2}} = \frac{\varepsilon}{2} < \varepsilon$$

$$\Rightarrow z \in B_\varepsilon(x)$$

but $z \notin B$ as $|z_n| = \frac{\varepsilon}{2}$ is not less than $\frac{1}{n}$

$$\Rightarrow \forall \varepsilon > 0, B_\varepsilon(x) \cap B^c \neq \emptyset$$

hence B is not open

(c) $(\ell_2, \|\cdot\|_2)$ is complete and $A \subset \ell_2$ is closed $\Rightarrow A$ is complete.

($A \subset M$ where M is complete and A is closed implies A is complete)

..... Rough Work

coordinate wise convergence.

$$d(x^{(k)}, x) \rightarrow 0$$

$$|x_i^{(k)} - x_i| \rightarrow 0$$

$$|x_i| \leq |x_i^{(k)} - x_i| + |x_i^{(k)}|$$

$$|x_i| \leq |x_i^{(k)} - x_i| + |x_i^{(k)}| \leq |x_i^{(k)} - x_i| + \frac{1}{k}$$

$$|x_i| \leq \frac{1}{i}$$

limit $k \rightarrow \infty$

$$|x_i| \leq \frac{1}{i}$$

set is not open.

$\forall \varepsilon > 0$.

$$B_\varepsilon(x) \cap$$

by archim

$$\frac{1}{n} > \frac{1}{\varepsilon}$$

$$A \text{ is } B_\varepsilon(x) \cap \bigcup_{i=1}^n B_{\frac{\varepsilon}{2}}(x_i)$$

$$\frac{1}{\varepsilon} \cdot y \in B_{\frac{\varepsilon}{2}}(x) \text{ \& } y \in B_{\frac{\varepsilon}{2}}(x_i)$$

$$x \in B_\varepsilon(y) \text{ also } B_{\frac{\varepsilon}{2}}(x_i) \subset B_\varepsilon(y)$$

consider $a \in A$

$$A \quad d(x, a) < \varepsilon$$